Forecast uncertainty and simulated storage scheduling

Local technical revision v1.1.0 · 18 October 2026

Evidence scope. This report documents the reference package prepared on 18 October 2026. Historical personal-run and CI records retain their original dates and version scope. The later local reproduction is recorded separately in PROJECT_HISTORY.md.

Abstract

A battery scheduling benchmark connects seven hourly forecast quantiles to risk-neutral and mean-CVaR decisions. This revision corrects an omitted physical restriction by adding binary charge/discharge modes, imports the corrected Project 1 forecasts, and uses common random numbers to compare uncertainty modes fairly. Across 83 simulated scheduling days, all saved corrected schedules satisfy full physical diagnostics at tolerance 1e-7. At CVaR weight .55, adaptive scenarios yield mean profit 8.34 EUR/day and empirical 90% CVaR loss 17.90 EUR, compared with raw scenarios at 9.82 and 20.67 EUR. Adaptive scenarios therefore sacrifice mean return for lower tail loss in this particular run; they do not dominate all alternatives. Three scenario-draw seeds and three simulated market seeds expose meaningful sensitivity.

Study design and information timing

Each test day contains 24 scheduling periods. The baseline input has 1,992 corrected P1 rows, 83 days, seven ordered quantiles, realised simulated prices, seasonal forecasts, and static/adaptive interval endpoints. Each stochastic strategy receives 60 scenarios. The battery has 2 MWh capacity, 1 MW power, charging and discharging efficiency .94, initial/terminal SOC 1 MWh, and throughput cost 2 EUR/MWh. The CVaR confidence level is .90. Forecast fitting and calibration are documented in the corrected P1 report.

Horizon 1 has issue time equal to target time on every test day. No prior-day forecast publication or market gate-closure timestamp is supplied. The experiment is consequently a **simulated 24-hour scheduling benchmark**; it does not establish operational day-ahead availability, implement bid curves or measure realised market revenue.

Strategy, weight .55	Mean profit	5th percentile	CVaR loss	Mean regret
Seasonal naive	-28.28	-80.73	82.64	84.93
Median risk neutral	8.69	-22.75	24.00	47.95
Raw CVaR	9.82	-15.23	20.67	46.82
Static CVaR	8.31	-13.54	18.46	48.33
Adaptive risk neutral	7.63	-21.95	27.39	49.02
Adaptive CVaR	8.34	-14.50	17.90	48.30
Perfect-foresight oracle	56.65	26.24	-24.68	0.00

All amounts are simulated EUR/day except the percentile and CVaR, which are EUR over daily outcomes. Regret is oracle profit minus strategy profit. A negative CVaR loss means that even the worst fixed tail mass remains profitable.

Physical model and the correction

For period t, charge c_t and discharge d_t are non-negative MW, SOC e_t is MWh, and mode z_t is binary. With one-hour periods, the model enforces:

0 <= c_t <= P*z_t; 0 <= d_t <= P*(1-z_t); z_t in {0,1}

e_(t+1) = e_t + eta_c*c_t - d_t/eta_d; 0 <= e_t <= E

e_0 = e_T = E*initial_soc_fraction

Each scenario s gives profit pi_s = sum_t[p_(s,t)*(d_t-c_t) - k*(c_t+d_t)]. All scenarios share one schedule. The minimisation objective is negative mean scenario profit plus weight * CVaR_alpha(-pi). SciPy's milp interface calls HiGHS with relative MIP gap 1e-9. Binary modes enforce exclusivity inside the optimization; solver schedules are not clipped, cancelled or otherwise repaired after solving. SciPy documents the integrality and bound interface used here [2,3].

Why exclusivity cannot be assumed

At a single price of -100 EUR/MWh, a lossy simultaneous cycle can be profitable in the continuous model while restoring terminal SOC. The reproducible regression uses E=2, P=1, both efficiencies .94, initial/terminal SOC=1 and cost=2:

Formulation	Charge MW	Discharge MW	End SOC MWh	Profit EUR
Historical continuous LP	1.0000	.8836	1.0000	7.8728
Corrected binary MILP	.0000	.0000	1.0000	.0000

This is a physical model defect even if a particular prior reference run happened not to show simultaneous activity. The optional lp_relaxation formulation exposes the historical behavior for comparison; it is not the default schedule.

Fixed-probability-mass CVaR retained

Rockafellar-Uryasev CVaR minimises zeta + mean(max(loss_s-zeta,0))/(1-alpha) [1]. The implementation retains the verified exact empirical fixed-mass estimator. Sort losses from worst to best and average exactly (1-alpha)*n observations, weighting the boundary observation fractionally. Thus 60 scenario losses use six observations, while 83 evaluation-day losses use 8.3. Ties must not expand that probability mass. Independent LP comparisons, fractional-mass and near-tie tests verify the estimator.

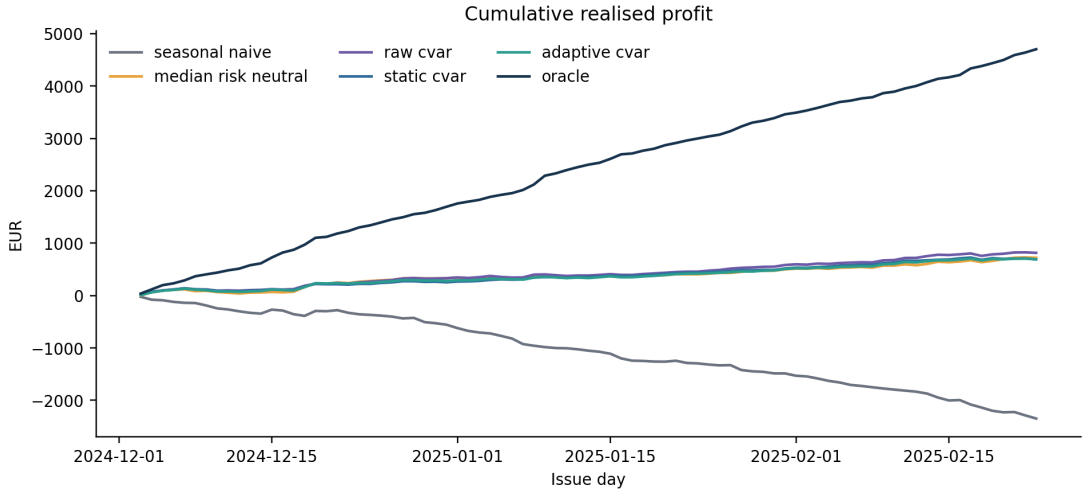
Full diagnostics check finite values, non-negativity, power bounds, all SOC endpoints, every SOC transition and inter-period continuity, initial/terminal SOC and exclusive operation. Counts are reported separately and in aggregate; tolerance is 1e-7.

Scenario construction and comparative results

Scenario uniforms use one shared normal factor and independent hourly factors: latent = .65*common + sqrt(1-.65^2)*idiosyncratic. Distinct latent hours therefore have Gaussian Pearson correlation .4225. This exchangeable dependence structure is a modeling assumption. It does not estimate hour-specific temporal dynamics.

Each hour interpolates the seven forecast quantiles at the latent rank. Values beyond q05/q95 use the boundary quantiles. Static and adaptive curves rescale deviations about q50 by clip(interval_width / max(q90-q10,1e-6), .35, 4). This is a width-based heuristic, not the transformation of conformal coverage into a calibrated joint price distribution. It need not reproduce asymmetric interval endpoints. In the corrected baseline, the upper cap affects one static and three adaptive rows; no source interval is infinite. Other valid P1 outputs may have infinite intervals when a finite-sample order is unavailable: P2 explicitly counts them and caps their scale at four, discarding interval equivalence.

For each day, the seed is scenario_seed + 17*day_index. Raw, static and adaptive curves use identical uniforms. All frontier weights reuse the same draws as the main adaptive strategy. Thus changing an uncertainty mode or weight does not introduce an unrelated random sample.



Cumulative simulated profits, baseline scenario seed 20260817.

CVaR weight	Raw mean / tail	Static mean / tail	Adaptive mean / tail
.55	9.82 / 20.67	8.31 / 18.46	8.34 / 17.90
.70	9.63 / 20.12	8.19 / 18.37	8.17 / 17.95
.85	9.51 / 19.67	8.64 / 16.86	8.41 / 16.62

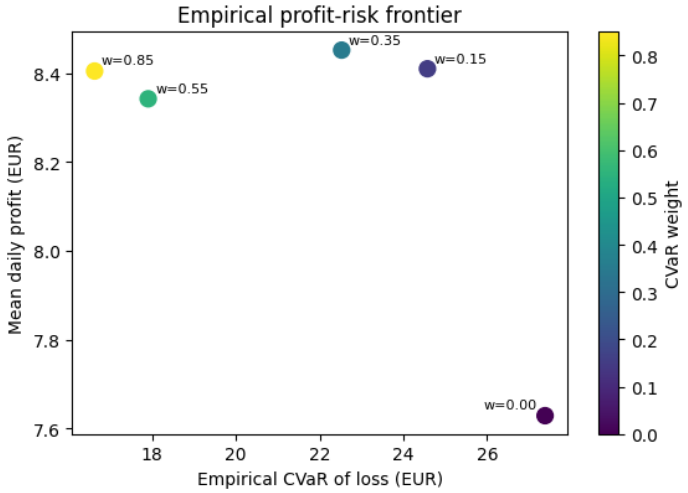
Cells contain mean daily profit / empirical CVaR loss in EUR. Observed out-of-sample profit and tail loss need not vary monotonically with the in-sample penalty. The raw strategy has higher mean profit in all three baseline comparisons; the adaptive strategy has lower tail loss. The oracle has future information and is only a diagnostic upper bound.

Sensitivity to finite scenario draws

Three scenario-draw seeds (20260817, 20260818, 20260819) use the same corrected baseline P1 forecasts and market outcomes. Each run uses 60 scenarios and 83 full scheduling days. These are Monte Carlo sensitivity checks, not three independent market backtests. Numbers below are averages and ranges over these three seeds; they are not confidence intervals.

Weight	Strategy	Mean profit avg	Profit min-max	CVaR loss avg	Tail min-max
.55	Raw	9.11	7.79-9.82	23.24	20.67-26.90
.55	Static	8.25	7.41-9.04	19.81	17.61-23.34
.55	Adaptive	8.48	7.48-9.60	19.59	17.70-23.18
.70	Raw	9.10	7.92-9.74	22.43	20.12-24.92
.70	Static	8.36	7.50-9.38	17.70	15.30-19.45
.70	Adaptive	8.56	7.94-9.55	18.53	16.49-21.17
.85	Raw	9.13	7.98-9.91	21.57	19.67-23.19
.85	Static	8.55	7.77-9.24	16.13	13.87-17.65
.85	Adaptive	8.61	8.01-9.42	17.64	15.68-20.60

The adaptive strategy has lower average tail loss than raw across these seeds, but static has lower average tail loss than adaptive at weights .70 and .85. Hence the study supports a conditional risk-return tradeoff, not universal adaptive superiority. The spread across just three draws also motivates larger scenario sets or explicit scenario-count convergence checks before stronger conclusions.



Adaptive risk frontier on the common baseline scenario draws.

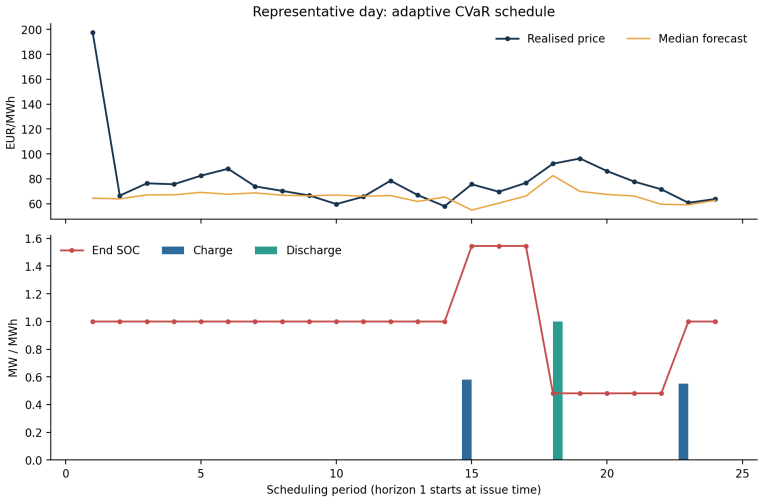
The plotted points describe a discrete empirical evaluation curve, not a certified efficient frontier. Mean profit rises and falls because realized benchmark outcomes differ from the scenario objective. Right-edge labels are placed inward so the colorbar cannot obscure them.

Sensitivity across three simulated market seeds

Two additional corrected P1 exports regenerate the simulated market and refit the forecast/calibration pipeline with seeds 20260818 and 20260819. They each provide 1,992 rows for the same 83 test-day positions. P2 keeps scenario seed 20260817, CVaR weight .55, 60 scenarios and identical battery settings. These checks vary the simulated data realization; they do not substitute for independent real-market validation.

Market seed	Raw mean / tail	Static mean / tail	Adaptive mean / tail
20260817	9.82 / 20.67	8.31 / 18.46	8.34 / 17.90
20260818	10.22 / 25.89	8.71 / 19.80	8.78 / 20.53
20260819	11.57 / 21.29	10.80 / 18.24	11.04 / 17.11

Cells contain mean daily profit / CVaR loss in EUR. Adaptive has lower observed tail loss than raw on all three simulated market seeds. Static has lower tail loss than adaptive on seed 20260818, while raw has higher mean profit on each seed. Three synthetic realizations do not establish statistical significance or generalization across electricity markets.



Representative exclusive adaptive CVaR schedule in the baseline simulated market.

Historical comparisons require care

The old v1.0.2 baseline reported adaptive mean profit 9.8587 EUR/day and CVaR loss 16.2264 EUR. These are historical results, not current values. The corrected baseline is 8.3429 and 17.9035 EUR. The input P1 CQR correction, common random numbers and MILP formulation changed together, so their difference is not an estimate of the isolated effect of exclusivity. Prior personal checks reproduced the old software and do not validate the corrected edition.

The complete historical input, numerical outputs, prior report, notes and report-generation script are retained under historical_v102/. No old evidence has been rewritten as new verification.

Verification, boundaries and reproducibility

The local Windows run used Python 3.13.15, NumPy 2.3.5, pandas 2.2.3 and SciPy 1.17.0 from the existing read-only environment. Imports were explicitly checked to come from this revised repository's src/storage_decision/. Bytecode writing was disabled and Matplotlib's cache was redirected inside the local workspace.

The final unit-test log records 20 passing tests, including the negative-price regression, full constraint diagnostic checks, finite/empty input and configuration validation, common-rank sampling, infinite interval handling and empirical CVaR LP equivalence. The release verifier independently reran weights .55/.70/.85 and passed 18/18 table comparisons at absolute tolerance 1e-8 with maximum numeric difference zero. It recalculated complete diagnostics from 3,486 saved/rerun daily schedules (83,664 scheduling periods). Manifests certify v1.1.0, binary MILP, common random numbers, source hashes and the timing limitation. Historical CI passes do not apply to this revision, and no new remote run is claimed.

The additional scenario-draw study contains nine complete runs; the separate market-seed study reuses the baseline plus two new runs. Every run includes all daily strategies, full saved schedules, frontier and battery sensitivity. A separate audit reloaded all 11 unique saved runs: 6,391 daily schedules and 153,384 periods, zero violations, maximum SOC-transition residual 5.9952e-14 MWh and zero terminal-SOC error. Floating-point residuals near machine precision are retained without postprocessing.

Practical limits

The simulator has no observed grid or market data. Its seasonal patterns, regime changes, price tails and dependence choices may differ materially from a real market. Marginal forecast intervals alone do not establish joint price-path calibration; clipping tails and widths further prevents a conformal coverage interpretation for the generated scenarios. The model assumes a price-taking battery, perfect execution and hourly settlement, with a linear throughput cost. It omits network limits, bid formation, imbalance, reserve products, gate closure and detailed degradation. A deployable extension would first establish publication-time-safe real inputs and validate the entire scenario/decision pipeline on independent periods.

Reproduction and evidence

Follow the README to create fresh local_runs/ outputs; do not overwrite supplied references. Study scripts reuse existing cached runs; the release verifier independently reruns the three weights. Current results are under corrected_v110/; outputs/ mirrors the baseline. Logs are in evidence/local_revision_20261018/. NEW_RELEASE_SUMMARY.md indexes checks. Baseline input SHA256: 6901c84f413ba1375fd8096c3c4910c2346d479a7ff18274832c2d27aba8472b.

References

[1] Rockafellar, R. T. and Uryasev, S. (2000). Optimization of Conditional Value-at-Risk. *Journal of Risk*, 2(3), 21-41. Original author-hosted paper: https://sites.math.washington.edu/~rtr/papers/rtr179-CVaR1.pdf

[2] SciPy. scipy.optimize.milp: mixed-integer linear programming, integrality/bounds/constraints and HiGHS interface. Official documentation: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.milp.html (consulted 18 October 2026; local experiments used SciPy 1.17.0).

[3] HiGHS. Official project documentation and solver description: https://highs.dev/ (consulted 18 October 2026).